



MATERIAL SAFETY DATA SHEET

1. Chemical product and company identification

Important information	*** This Safety Data Sheet is only authorised for use by HP for HP Original products. Any unauthorised use of this Safety Data Sheet is strictly prohibited and may result in legal action being taken by HP. ***		
A. Product name	1XA54Series		
Other means of identification	Not available.		
B. Recommended use and Limitations on use			
Recommended use	Inkjet printing		
C. Supplier information			
	HP Korea Inc. 07325 83 Uisadang-daero Youngdeungpo-gu Seoul, Korea		
Telephone	02-780-6200		
HP Inc. health effects line			
(Toll-free within the US)	1-800-457-4209		
(Direct)	1-760-710-0048		
HP Inc. Customer Care Line			
(Toll-free within the US)	1-800-474-6836		
(Direct)	1-208-323-2551		
Email:	sustainability@hp.com		

2. Hazards identification

A. Hazard category/Classification

Physical hazards	Not classified.	
Health hazards	Sensitization, skin	Category 1
Environmental hazards	Not classified.	

B. Warning label items including precautionary statement

• Pictogram



• Signal word	Warning
• Hazard statement	None.

• Precautionary statement

Prevention

P261	Avoid breathing dust/fume/gas/mist/vapors/spray.
P272	Contaminated work clothing should not be allowed out of the workplace.
P280	Wear protective gloves/protective clothing/eye protection/face protection.

Response

P302 + P352	IF ON SKIN: Wash with plenty of water.
P333 + P313	If skin irritation or rash occurs: Get medical advice/attention.
P321	Specific treatment (see this label).
P362 + P364	Take off contaminated clothing and wash it before reuse.

Disposal

P501	Dispose of contents/container in accordance with local/regional/national/international regulations.
------	---

C. Other hazards not included in the hazard category criteria (e.g. dust explosion hazard) Complete toxicity data are not available for this specific formulation. Potential routes of overexposure to this product are skin and eye contact. Inhalation of vapor and ingestion are not expected to be significant routes of exposure for this product under normal use conditions.

Supplemental information None.

3. Composition/information on ingredients

Chemical identity	Common and alternative names	CAS number	ID number	Content in percent (%)
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)		2634-33-5	KE-02680	0.01-0.1
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)		2682-20-4	KE-24316, 2012-1-645	0.01-0.1

Composition comments This ink supply contains an aqueous ink formulation.

4. First aid measures

- A. In case of eye contact** Do not rub eyes. Immediately flush with large amounts of clean, warm water (low pressure) for at least 15 minutes or until particles are removed. If irritation persists get medical attention.
- B. In case of skin contact** Wash affected areas thoroughly with mild soap and water. Get medical attention if irritation develops or persists.
- C. In case of inhalation** Remove to fresh air. If symptoms persist, get medical attention.
- D. In case of swallowing** If ingestion of a large amount does occur, seek medical attention.
- E. Note to physician** Not available.

5. Fire-fighting measures

- A. Suitable (and unsuitable) extinguishing media**
- Suitable extinguishing media** Dry chemical, CO₂, water spray or regular foam.
- Unsuitable extinguishing media** None known.
- B. Specific hazards arising from the chemical (example: hazardous combustion products)** Not available.
- C. Specific methods of fire-fighting** Not available.
- Specific methods** None established.

6. Accidental release measures

- A. Personal precautions, protective equipment and emergency measures** Wear appropriate personal protective equipment.
- B. Environmental precautions** Do not let product enter drains. Do not flush into surface water or sanitary sewer system.
- C. Methods and materials for containment and cleaning up** Dike the spilled material, where this is possible. Absorb with inert absorbent such as dry clay, sand or diatomaceous earth, commercial sorbents, or recover using pumps.

7. Handling and storage

- A. Precautions for safe handling** Avoid contact with skin, eyes and clothing.
- B. Conditions for safe storage (including any incompatibilities)** Keep out of the reach of children. Keep away from excessive heat or cold.

8. Exposure controls/personal protection

- A. Exposure limit values, biological limit values, etc**
- Korea. Exposure Limits for Chemicals and Physical Agents, Occupational Safety and Health Act "K-OSHA" Article 106**
- | Components | Type | Value | Form |
|--|------|-----------------------|---------------------|
| 2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone) (CAS 2682-20-4) | TWA | 0.1 mg/m ³ | Inhalable fraction. |

Biological limit values	No biological exposure limits noted for the ingredient(s).
Exposure guidelines	Exposure limits have not been established for this product.
B. Appropriate engineering controls	Not available.
C. Personal protective equipment	
• Respiratory protection	Not available.
• Eye protection	Wear safety glasses with side shields (or goggles).
• Hand protection	Use protective gloves made of: Nitrile rubber.
• Body protection	Wear appropriate chemical resistant clothing.
Hygiene measures	Handle in accordance with good industrial hygiene and safety practice.

9. Physical and chemical properties

A. Appearance	
Physical state	Liquid.
Form	Not available.
Color	Magenta
B. Odor	Not available.
C. Odor threshold	Not available.
D. pH	8 - 10
E. Melting point/freezing point	Not available.
F. Boiling point, initial boiling point, and boiling range	Not available.
G. Flash point	>230.0 °F (>110.0 °C)
H. Evaporation rate	Not available.
I. Flammability (solid, gas)	Not available.
J. Upper/lower limit on flammability or explosive limits	
Explosive limit - lower (%)	Not available.
Explosive limit - upper (%)	Not available.
K. Vapor pressure	Not available.
L. Solubility	
Solubility (water)	Not available.
M. Vapor density	Not available.
N. Specific gravity	1 - 1.1
O. n-octanol/water partition coefficient	Not available.
P. Auto-ignition temperature	Not available.
Q. Decomposition temperature	Not available.
R. Viscosity	Not available.
S. Molecular weight	Not available.
Other data	
Density	1.05 g/cm3
Flammability	Not flammable according to GHS Hazard Classification Criteria.
Kinematic viscosity	<3 mm²/s at 25°C (calculated).
Oxidizing properties	Not determined
Particle size	Not applicable.
VOC	<152 g/l
9.2. Other information	
Density	1.05 g/cm3

10. Stability and reactivity

A. Stability and hazardous reaction potential	
Stability	Stable under recommended storage conditions.

Hazardous reaction potential	None known.
B. Conditions to avoid (e.g. static discharge, shock or vibration, etc)	Not available.
C. Incompatible materials	Incompatible with strong bases and oxidizing agents.
D. Hazardous decomposition products	Upon decomposition, this product may yield gaseous nitrogen oxides, carbon monoxide, carbon dioxide and/or low molecular weight hydrocarbons., fluorinated hydrocarbons and hydrogen fluoride.

11. Toxicological information

A. Information on likely routes of exposure

- **Respiratory organs** Under normal conditions of intended use, this material is not expected to be an inhalation hazard.
- **Skin** May cause an allergic skin reaction.
- **Eyes** Contact with eyes may result in mild irritation.
- **Mouth** Ingestion is not a likely route of exposure.

B. Information on health hazards

- **Acute toxicity (list all possible routes of exposure)** Based on available data, the classification criteria are not met.

Components	Species	Test Results
1,2-Benzisothiazolin-3-one (Benzisothiazolinone) (CAS 2634-33-5)		
Acute		
Dermal		
LD50	Rat	> 2000 mg/kg (OECD 402)
Oral		
LD50	Mouse	1150 mg/kg
	Rat	1020 mg/kg
		670 mg/kg (OECD 401)

2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone) (CAS 2682-20-4)

Acute		
Dermal		
LD50	Rat	242 mg/kg (OECD 402)
Inhalation		
LC50	Rat	0.11 mg/l, 4 h (OECD 403)
Oral		
LD50	Rat	120 mg/kg

- **Corrosivity or irritation to the skin** Non irritant in rabbit (OECD 404) Based on available data, the classification criteria are not met.

• Irritation Corrosion - Skin

2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	Corrosive, rabbit (OECD 404)
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	Irritating (4 h, rabbit)

- **Serious eye damage/eye irritation** Based on available data, the classification criteria are not met.

Eye	
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	Causes serious eye damage (rabbit)
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	Corrosive, based on OECD 404 results

- **Respiratory sensitization** Based on available data, the classification criteria are not met.

- **Skin sensitization** May cause an allergic skin reaction.

Skin sensitization	
1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	Causes sensitization (Guinea pig, OECD 406)
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	Sensitizing, mice (OECD 429), Sensitizing, guinea pigs (OECD 406)

• Carcinogenic properties /Carcinogenicity	Based on available data, the classification criteria are not met.
• Mutagenic properties /Mutagenicity	Based on available data, the classification criteria are not met.
• Reproductive toxicity	Based on available data, the classification criteria are not met.
• Specific target organ toxicity - single exposure	Based on available data, the classification criteria are not met.
• Specific target organ toxicity - repeated exposure	Based on available data, the classification criteria are not met.
• Aspiration hazard	Based on available data, the classification criteria are not met.

12. Ecological information

A. Ecotoxicity No information available.

Product		Species	Test Results
1XA54Series			
Aquatic			
Fish	LC50	Fathead minnow (Pimephales promelas)	> 750 mg/l, 96 Hours
Components		Species	Test Results
1,2-Benzisothiazolin-3-one (Benzisothiazolinone) (CAS 2634-33-5)			
<i>Acute</i>			
	EC50	Activated sludge	12.8 mg/l, 3 h (OECD 209)
Other	EC50	Pseudokirchnerella subcapitata	0.11 mg/l, 72 h OECD (201)
	NOEC	Pseudokirchnerella subcapitata	0.055 mg/l, 72 h (OECD 201)
Aquatic			
<i>Acute</i>			
Crustacea	EC50	Daphnia magna	4.4 mg/l, 48 h 2.9 mg/l, 48 h (OECD 202)
Fish	LC50	Oncorhynchus mykiss	2.15 mg/l, 96 h (OECD 203) 0.8 mg/l, 96 h
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone) (CAS 2682-20-4)			
<i>Acute</i>			
	EC50	Activated sludge	34.6 mg/l (DIN 38412-3)
Other	EC50	Pseudokirchnerella subcapitata	0.445 mg/l, 120 h (OECD 201)
Aquatic			
<i>Acute</i>			
Crustacea	EC50	Daphnia magna	1.68 mg/l, 48 h (OECD 202)
Fish	LC50	Rainbow Trout	6 mg/l, 96 h (OECD 203)
<i>Chronic</i>			
Crustacea	NOEC	Daphnia magna	0.0442 mg/l, 21 d (OECD 211)
Fish	NOEC	Oncorhynchus mykiss	4.93 mg/l, 98 d (OECD 210)

Aquatic toxicity No information available.

B. Persistence/degradability

Biodegradability

Percent degradation (Aerobic biodegradation-ready)

1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	85 %, Not readily biodegradable (OECD 301C) Test Duration: 63 d
2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)	54.1 %, (OECD 301B) Test Duration: 29 d

C. Bioaccumulative potential

Bioconcentration factor

1,2-Benzisothiazolin-3-one (Benzisothiazolinone)	6.62, (OECD 305) Species: Bluegill (Lepomis macrochirus)
--	---

Bioconcentration factor

2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)

48.1, Viscera (1972)

Species: Bluegill (Lepomis macrochirus)

5.75, Carcass (1972)

Species: Bluegill (Lepomis macrochirus)

Octanol/water partition coefficient log Kow

1,2-Benzisothiazolin-3-one (Benzisothiazolinone)

0.7 (OECD 117)

2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone)

-0.32 (OECD 107)

D. Mobility in soil

Not available.

Adsorption**Soil/sediment sorption - log Koc**

1,2-Benzisothiazolin-3-one (Benzisothiazolinone)

0.97, (OECD 121)

E. Other adverse effects

Not available.

13. Disposal considerations**A. Method of disposal**

Do not allow this material to drain into sewers/water supplies.

Dispose of waste material according to Local, State, Federal, and Provincial Environmental Regulations.

HP's Planet Partners (trademark) supplies recycling program enables simple, convenient recycling of HP original inkjet and LaserJet supplies. For more information and to determine if this service is available in your location, please visit <http://www.hp.com/recycle>.**B. Disposal considerations
(including disposal of
contaminated containers or
packaging)**

Not available.

14. Transport information**DOT****UN number**

Not available.

UN proper shipping name

Not Regulated

Transport hazard class(es)**Class**

Not available.

Subsidiary hazard

-

Packing group

Not available.

Environmental hazards**Marine pollutant**

No

Special precautions for user

Not available.

IATA**UN number**

Not available.

UN proper shipping name

Not Regulated

Transport hazard class(es)**Class**

Not available.

Subsidiary hazard

-

Packing group

Not available.

Environmental hazards

No

Special precautions for user

Not available.

IMDG**UN number**

Not available.

UN proper shipping name

Not Regulated

Transport hazard class(es)**Class**

Not available.

Subsidiary hazard

-

Packing group

Not available.

Transport hazard class(es)**Marine pollutant**

No

EmS

Not available.

Special precautions for user

Not available.

ADR**UN number**

Not available.

UN proper shipping name

Not Regulated

Transport hazard class(es)**Class**

Not available.

Subsidiary hazard	-
Hazard No. (ADR)	Not available.
Tunnel restriction code	Not available.
Packing group	Not available.
Environmental hazards	No
Special precautions for user	Not available.
Further information	Not a dangerous good under DOT, IATA, ADR, IMDG, or RID.

Transport in bulk according to Annex II of MARPOL73/78 and the IBC code: Not applicable.

15. Regulatory information

A. Restrictions under the Industrial Safety and Health Law

Harmful Substances Prohibited from Manufacturing

Not regulated.

Harmful Substances Requiring Permission for Manufacture or Use

Not regulated.

Controlled Hazardous Substances

Not regulated.

Harmful Substances Requiring Special Medical Examination

Not regulated.

Workplace Environmental Monitoring Harmful Materials

Not regulated.

Occupational Exposure Limit

2-Methyl-2h-isothiazol-3-one (Methylisothiazolinone) (CAS 2682-20-4)

B. Restrictions under the Chemicals Control Law

Accidental Release Not regulated.

Prevention Substances

C. Act on the Registration and Evaluation of Chemicals

Hazardous Chemical Not regulated.

Substances Acutely Hazardous to Human Health

Hazardous Chemical Not regulated.

Substances - Chronically Hazardous to Human Health

Hazardous Chemical Not regulated.

Substances - Hazardous to the Environment

Substance with Unknown Not regulated.

Hazards

Restricted/Prohibited/Permitted Chemical Substances Not regulated.

Intensive Control Not regulated.

Substances

Priority Existing Chemicals (PECs) Not regulated.

D. Restrictions under the Wastes Control Act

Halogenated Materials in Waste Organic Solvents

Not regulated.

Hazardous Substances

Not regulated.

Other information Contains a short-chain, partially fluorinated substituted glycol at < 0.1%.

E. Restrictions under other foreign or domestic laws

Clean Air Conservation Act

Air Pollutants

Not regulated.

Not listed.

Specific Air Pollutants

Not regulated.

Regulatory information

HP complies with chemical regulatory requirements in chemical substance notification laws, where applicable. All chemical substances are notified or exempt from notification or listed in the inventory as existing substances in the following countries: US (TSCA), Canada (DSL/NDSL), Australia (AIC), Japan (ISHL, ENCS), Philippines (PICCS), New Zealand (NZIoC), Russia and China (IECSC). For guidance on importation and/or additional requirements for registration schemes such as EAEU, EU, South Korea, Turkey, UK, India and Taiwan, please contact the Sustainability and Compliance Center (sustainability@hp.com).

16. Other information

- A. Source of information** Not available.
- B. Issue date** 16-Oct-2018
- C. Number of revisions and date of most recent revision** 08-Jun-2026 (11 revision)
- D. Other** Not available.

Disclaimer

This Safety Data Sheet document is provided without charge to customers of HP. Data is the most current known to HP at the time of preparation of this document and is believed to be accurate. It should not be construed as guaranteeing specific properties of the products as described or suitability for a particular application. This document was prepared to the requirements of the jurisdiction specified in Section 1 above and may not meet regulatory requirements in other countries.

This safety data sheet is meant to convey information about HP inks (toners) provided in HP Original ink (toner) supplies. If our Safety Data Sheet has been provided to you with a refilled, remanufactured, compatible or other non-HP Original supply please be aware that the information contained herein was not meant to convey information about such products and there could be considerable differences from information in this document and the safety information for the product you purchased. Please contact the seller of the refilled, remanufactured or compatible supplies for applicable information, including information on personal protective equipment, exposure risks and safe handling guidance. HP does not accept refilled, remanufactured or compatible supplies in our recycling programs.

Explanation of abbreviations

ACGIH	American Conference of Governmental Industrial Hygienists
Acute Tox.	Acute toxicity
Aquatic Acute	Short-term (acute) aquatic hazard
Aquatic Chronic	Long-term (chronic) aquatic hazard
Asp. Tox.	Aspiration hazard
Carc.	Carcinogenicity
CAS	Chemical Abstracts Service
CERCLA	Comprehensive Environmental Response Compensation and Liability Act
CFR	Code of Federal Regulations
COC	Cleveland Open Cup
DOT	Department of Transportation
EPCRA	Emergency Planning and Community Right-to-Know Act (aka SARA)
Eye Dam.	Serious eye damage
Eye Irrit.	Eye Irritation
Flam. Liq.	Flammable liquids
Flam. Sol.	Flammable solids
Lact.	Effects on or via lactation
Muta.	Germ cell mutagenicity
IARC	International Agency for Research on Cancer
NIOSH	National Institute for Occupational Safety and Health
NTP	National Toxicology Program
OSHA	Occupational Safety and Health Administration
Ox. Liq.	Oxidising liquids
Ozone	Hazardous to the ozone layer
PEL	Permissible Exposure Limit
Press. Gas	Gases under pressure
RCRA	Resource Conservation and Recovery Act
REC	Recommended
REL	Recommended Exposure Limit
Repr.	Reproductive toxicity
Resp. Sens.	Respiratory sensitization
SARA	Superfund Amendments and Reauthorization Act of 1986
Skin Corr.	Skin corrosion
Skin Irrit.	Skin irritation
Skin Sens.	Skin sensitization
STEL	Short-Term Exposure Limit
STOT RE	Specific target organ toxicity - repeated exposure
STOT SE	Specific target organ toxicity - single exposure
TCLP	Toxicity Characteristics Leaching Procedure
TLV	Threshold Limit Value
TSCA	Toxic Substances Control Act